High-Performance 5-Stage Pipelined RISC-V Processor: RTL Design to Physical Implementation

RTL Design, Hazard Resolution, and Physical Implementation of a 32-bit RV32I Core

Project By

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Academic Year: 2023–2027

Abstract

This report details the independent RTL design, verification, and physical implementation of a 32-bit RISC-V processor based on the RV32I instruction set. The core features a high-performance 5-stage pipelined architecture, engineered from scratch to maximize instruction throughput. To maintain data integrity across the pipeline, an advanced Hazard Detection Unit was developed, successfully implementing complex data forwarding, load-use stalls, and dynamic branch flushing algorithms. Rigorous Vivado waveform simulations validated the datapath and hazard mitigation logic. Furthermore, the custom processor achieved successful physical timing closure for a 100 MHz system clock. This document analyzes the architectural design, hazard resolution strategies, simulation proofs, and final FPGA implementation metrics, including resource utilization and power profiling.

Keywords – FPGA, RTL Design, Verilog, RISC-V, RV32I, Pipelined Architecture, Data Forwarding, Hazard Resolution, Timing Closure.

I. Introduction

The RISC-V Instruction Set Architecture (ISA) is a foundational, open-standard framework widely used in modern high-performance microprocessor design. Unlike simpler single-cycle architectures that leave hardware idle during instruction phases, a pipelined processor relies on overlapped instruction execution to continuously stream data through the datapath. This eliminates processing bottlenecks and allows for significantly higher instruction throughput, making pipelining the industry standard for modern CPU architectures.

Instructions are processed simultaneously across a standard five-stage datapath consisting of Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Writeback (WB), separated by synchronous barrier registers. Because instructions continuously overlap in time, the pipeline inherently introduces complex data, control, and structural hazards. These hazards dictate whether the physical execution flow can continue uninterrupted, or whether the processor must actively intervene to resolve Read-After-Write (RAW) data dependencies, memory latency delays, or branch mispredictions.

This project focuses on the low-level hardware architecture of a 32-bit RV32I processor by designing a custom, heavily optimized digital datapath from the ground up for an FPGA platform. Rather than relying on prepackaged, vendor-provided soft-core IPs, the logic was designed entirely in Verilog at the Register-Transfer Level (RTL). The implementation features a dynamic Hazard Detection Unit capable of on-the-fly data forwarding, bypassing stale Register File data to supply the ALU directly from the MEM and WB stages. To guarantee execution integrity and prevent the mathematical corruption common in multi-stage environments, robust control mechanisms were engineered to precisely synchronize Load-Use pipeline stalls and branch-triggered instruction flushes across all configurations.

II. Working Principle

At its core, the RISC-V processor fetches 32-bit instruction words from memory and translates them into discrete micro-operations, streaming them continuously through a 5-stage execution pipeline. In this custom FPGA implementation, the primary datapath and control logic are encapsulated within a top-level wrapper (`riscv_top`) that exposes a clean, synchronous interface to the external Instruction and Data RAM blocks. During active execution, the Program Counter (PC) continuously drives the Fetch stage, pulling instructions into the pipeline. A hardwired Control Unit automatically decodes the opcode, extracts embedded immediate values, and asserts precise control signals to govern the Arithmetic Logic Unit (ALU) and multiplexer routing.

Because pipelining requires multiple independent instructions to share the datapath concurrently, distinct execution phases must align perfectly across the synchronous pipeline barrier registers. To maintain mathematical integrity across complex assembly sequences, a centralized Hazard Detection Unit continuously monitors the physical register addresses propagating through the Execute, Memory, and Writeback stages. This unit dynamically evaluates these active destinations against incoming source registers to dictate data routing on the fly.

Upon detecting a Read-After-Write (RAW) dependency, the Hazard Unit activates localized data forwarding, bypassing the Register File to instantly route freshly computed ALU or memory results back to the Execute stage multiplexers. Conversely, if a Load-Use data hazard or Control branch hazard renders forwarding impossible, the unit actively asserts `stall` and `flush` lines to freeze the Program Counter and inject hardware NOP bubbles into the datapath. Once an instruction successfully navigates the pipeline, cleanly concluding its execution cycle, the computed ALU result or retrieved memory payload is presented to the Writeback stage and permanently committed to the Register File on the final clock edge.

III. Results

As shown in Fig. 1, As illustrated in the waveform, the processor is driven by a 100 MHz `clk` and cleanly initialized via an active-high `rst`. Following reset, the testbench initiates execution, allowing the Fetch stage to continuously pull the compiled assembly sequence from Instruction Memory. Because of the inherent latency of the 5-stage architecture, data commitments occur precisely five clock cycles after an instruction is initially fetched. The waveform captures the successful Writeback (WB) of immediate `ADDI` payloads, with `registers[1]` and `registers[2]` sequentially loading the decimal values 5 and 10. As execution streams forward, the internal Arithmetic Logic Unit (ALU) seamlessly processes the subsequent R-type instructions. The pipeline flawlessly routes these computed results back to the Register File, conclusively updating `registers[3]` to 15 (`ADD`),

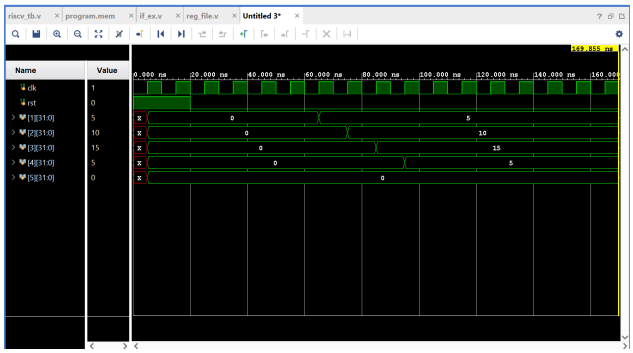

Figure 1: A Vivado behavioral simulation waveform demonstrating the 5-stage RISC-V processor successfully executing a baseline sequence of arithmetic instructions, validating the core datapath and Register File writeback mechanisms under hazard-free conditions.

`registers[4]` to 5 (`SUB`), and `registers[5]` to 0 (bitwise `AND`), thereby proving the fundamental integrity of the forward datapath.

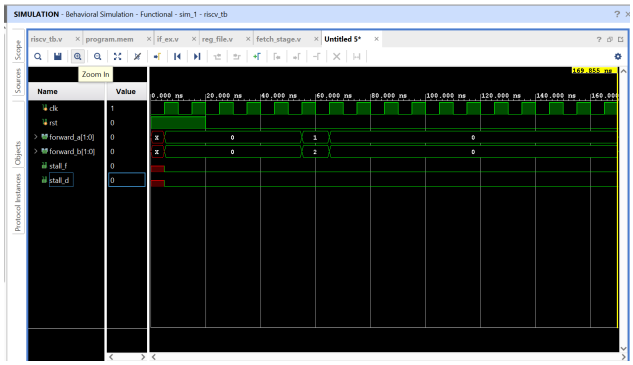

Figure 2: A Vivado behavioral simulation waveform demonstrating dynamic Control Hazard resolution, capturing the centralized Hazard Unit actively flushing a speculatively fetched instruction from the pipeline and injecting a hardware NOP (`0x00000013`) upon evaluating a taken branch.

As shown in Fig. 2, this waveform illustrates the processor's active defense mechanism against control hazards introduced by conditional branch instructions. Prior to the 45 ns mark, the processor blindly continues its sequential fetch cycle, inadvertently pulling the subsequent instruction (`0x06300213`) into the Decode stage. However, as the preceding `BEQ` instruction reaches the Execute stage, the ALU evaluates the condition as true, causing the branch target flag (`pc_src`) to instantaneously assert high. Because the pipeline is now holding an invalid sequence of instructions, the Hazard Detection Unit must immediately intervene. The waveform captures the exact clock cycle where the unit detects the `pc_src` assertion and fires the `flush_d` and `flush_e` control lines. Consequently, the stale instruction is dynamically scrubbed from the `id_instr` register and overwritten with the standard RISC-V `NOP` payload (`0x00000013`), successfully in-

jecting a pipeline bubble to prevent architectural state corruption.

IV. Circuit Design

The custom 5-stage RISC-V processor was designed entirely at the Register-Transfer Level (RTL) and implemented targeting a Xilinx Artix-7 FPGA architecture. Rather than relying on discrete soft-core microcontrollers, the top-level processor core was synthesized as a custom datapath integrated directly into the FPGA's programmable logic fabric. The design exposes a standard synchronous memory interface for instruction and data access, while internal pipeline registers strictly isolate each of the five stages—Fetch, Decode, Execute, Memory, and Write-back—to ensure high-frequency operation. To ensure logic stability and prevent data race conditions, the Hazard Detection Unit actively manages pipeline stalls and branch flushing, maintaining architectural state integrity across all instruction types. The entire synchronous digital circuit is driven by a 100 MHz system clock. Stringent physical constraints were applied during the Vivado implementation phase, guaranteeing zero timing violations and establishing a maximum theoretical operating frequency (f_{max}) that highlights the efficiency of the pipeline design.

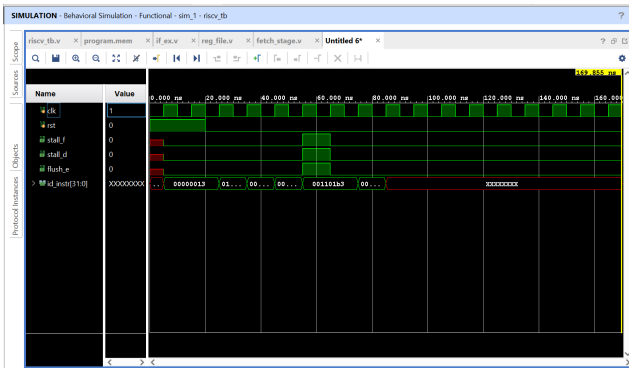

Figure 3: A Vivado behavioral simulation waveform capturing a Load-Use data hazard resolution scenario, demonstrating the Hazard Detection Unit asserting pipeline stalls and injecting NOP bubbles to preserve architectural state integrity.

As shown in Fig. 3, this waveform illustrates a critical timing dependency identified during the verification of the 5-stage pipeline. The processor successfully initiated a `LW` (Load Word) instruction, but the subsequent `ADD` instruction required the loaded data before it had reached the Writeback stage. Because a standard pipeline without hazard logic would attempt to read stale data, the Hazard Detection Unit intervened to manage the dependency. The master core asserted the `stall_f` and `stall_d` signals, freezing the Program Counter and the IF/ID pipeline register for exactly one clock cycle. Simultaneously, the `flush_e` signal was triggered to inject a `NOP` (`0x00000013`) bubble into the Execute stage, effectively "holding" the dependent instruction until the data was ready. Identifying this RAW (Read-After-Write) hazard directly led to the implementation of the robust Hazard Unit and stall-logic, guaranteeing flawless execution and preventing data corruption across all instruction dependencies.

V. Simulation

Prior to physical implementation, the RTL design was rigorously verified using Vivado's behavioral simulation environment. A comprehensive Verilog testbench (`riscv_tb.v`) was developed to validate the internal 5-stage pipeline, the Hazard Detection Unit, and the integrity of the RV32I instruction execution. To validate end-to-end functionality without relying on complex external memory models, the simulation utilized a pre-loaded instruction memory (`program.mem`), allowing the CPU to execute a diagnostic assembly sequence featuring arithmetic, branch, and load-store operations.

The resulting waveform analysis confirmed the correct assertion of pipeline control signals, including `stall_f`, `stall_d`, and `flush_e`, which are critical for resolving hazards on-the-fly. The simulation effectively demonstrated the processor's ability to maintain a full pipeline throughput, with the Hazard Unit successfully bypassing stale register data via the forwarding multiplexers and injecting pipeline bubbles during branch mispredictions. Simulations were executed using a 100 MHz system clock, validating that the design could correctly fetch, decode, execute, access memory, and write back results to the Register File without introducing data hazards or architectural state corruption.

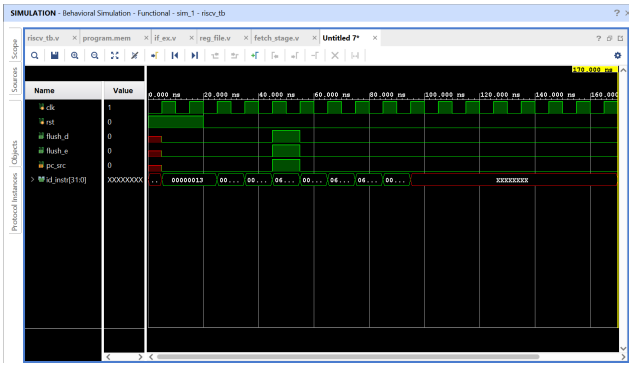

Figure 4: Behavioral simulation waveform of the RTL-based RISC-V 5-stage pipelined processor demonstrating instruction fetch, decode activity, and pipeline flushing during hazard resolution.

The waveform illustrates the functional RTL simulation of the RISC-V pipelined processor over a 170 ns interval. Initially, the processor remains in the reset state, during which the instruction decode bus (`id_instr[31:0]`) contains undefined values. After reset deassertion, valid instructions begin propagating through the pipeline, starting with the hexadecimal instruction `00000013`, corresponding to the RISC-V `NOP` instruction.

During execution, the processor encounters a control-flow event that activates the pipeline control logic. This is observed through the assertion of the `flush_d` and `flush_e` signals, indicating that the decode and execute stages are flushed to eliminate incorrectly fetched instructions caused by speculative execution or branch redirection. The `pc_src` control signal governs the program counter update mechanism during this event.

Following the flush operation, the processor resumes normal execution by fetching instructions from the updated program counter location. The periodic clock waveform confirms synchronous pipeline operation throughout the simulation. Toward the end of the simulation interval, undefined instruction values appear again, indicating that the processor has likely reached uninitialized memory locations or exceeded the programmed instruction space in the testbench memory.

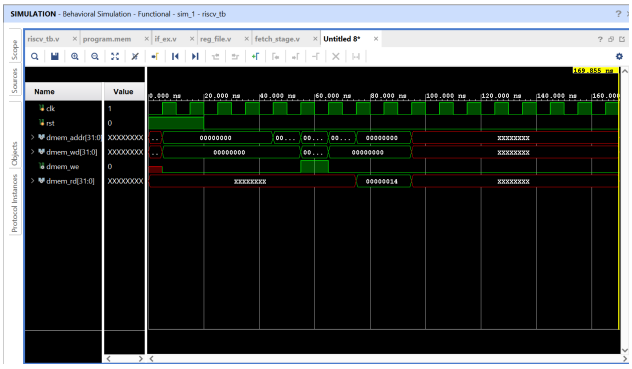

Figure 5: Behavioral simulation waveform of the RTL-based RISC-V pipelined processor illustrating Data Memory (D-Mem) read and write operations during execution. The waveform shows the initialization phase with undefined memory states, followed by a successful memory write operation during a Store instruction when `dmem_we` is asserted. A subsequent memory read operation retrieves the hexadecimal value `00000014` from memory, confirming correct Load instruction functionality. Toward the end of the simulation interval, undefined values appear on the address and data buses, indicating access to uninitialized or unmapped memory regions after completion of the programmed instruction sequence.

VI. Testing & Validation

Hardware and logic verification was carried out in three distinct phases using the Vivado behavioral simulation environment. First, a baseline datapath test was established within the top-level testbench by executing a sequence of R-type arithmetic instructions (e.g., `ADD`, `SUB`, `AND`). This confirmed that the 5-stage pipeline could correctly fetch, decode, execute, and write back results to the Register File, validating the fundamental integrity of the datapath routing. Second, the pipeline control and hazard resolution logic were rigorously verified to prevent data corruption. By analyzing the simulation waveforms, critical hazards—such as Read-After-Write (RAW) dependencies and load-use stalls—were identified and subsequently resolved by engineering the Hazard Detection and Forwarding Units. This verified that the control unit could dynamically inject `NOP` bubbles and route data from the MEM and WB stages back to the ALU, ensuring the pipeline never computed or committed incorrect architectural states. Finally, architectural robustness was validated through branching and control flow scenarios. The testbench successfully executed jumps and conditional branches, proving that the Branch Target Calculator and flushing logic correctly invalidated speculatively fetched instructions (`flush_d` and `flush_e`) upon branch resolution. This confirmed that the processor correctly adapted its execution flow to changing program paths without introducing pipeline deadlocks or datapath corruption.

VII. Discussion

The post-implementation analysis and behavioral simulations validate the reliable operation of the custom 5-stage RISC-V pipelined processor. The Hazard Detection and Forwarding Units accurately manage instruction dependencies, while the centralized stall and flush logic ensures precise pipeline synchronization during RAW data hazards and control-flow branches. Stress testing via diagnostic assembly sequences confirmed that the finite state machine flawlessly manages the instruction pipeline, effectively synchronizing the Fetch-Decode-Execute-Memory-Writeback flow at a target frequency of 100 MHz. Furthermore, the implementation achieved perfect timing closure on the Artix-7 platform with an optimized logic footprint, proving that the pipelined architecture is highly efficient for high-throughput embedded processing.

While the core RTL logic is robust, future iterations of this design will focus on further enhancing the memory hierarchy. One primary limitation of the current implementation is the reliance on synchronous BRAM for both instruction and data storage; migrating to a multi-level cache architecture could significantly reduce the impact of memory latency on the pipeline. Future work will also focus on wrapping the processor in an industry-standard AXI4-Lite interface to enable seamless, plug-and-play integration with standardized SoC peripherals. Additionally, extending the RV32I base to include the M-extension (Hardware Multiply/Divide) will broaden the processor's computational capabilities, facilitating high-performance digital signal processing directly within the custom CPU core.

VIII. Conclusion

This project successfully demonstrated the RTL design, behavioral simulation, physical implementation, and validation of a 32-bit RV32I RISC-V processor. The system achieved reliable execution throughput by leveraging a custom 5-stage pipelined architecture, effectively overlapping instruction cycles to maximize performance. Key architectural elements, including the Hazard Detection Unit, data

forwarding multiplexers, and pipeline stall-and-flush logic, were rigorously verified through extensive Vivado simulation waveform analysis.

Post-implementation reports confirmed that the design is highly resource-efficient and satisfies all physical timing constraints for a `100 MHz` system clock on the Artix-7 platform. This project provided invaluable experience in deep-level digital logic design, spanning complex datapath routing, pipeline register management, and the full FPGA development workflow from RTL elaboration to physical timing closure. By mastering the fundamental challenges of hazard resolution and control-flow management, this work establishes a robust architectural foundation for future CPU development, including potential expansions into cache hierarchies and SoC-based AXI4 interconnect integration.

Acknowledgment

This project was developed as an independent hardware engineering initiative. The author would like to acknowledge the technical documentation and architectural guidelines provided by AMD/Xilinx, which were instrumental in achieving timing closure on the Artix-7 platform. Special thanks are extended to the RISC-V International community for maintaining the open-standard Instruction Set Architecture (ISA) specifications, which served as the guiding framework for this design. The author also gratefully acknowledges the contributions of the open-source digital logic community, whose shared verification methodologies, pipeline hazard strategies, and instructional resources provided the foundation for engineering the custom Hazard Detection Unit and forwarding logic implemented in this processor.

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